From Narrative to Model: Formalizing, Testing and Simulating Assigned Scenarios with Temporal Knowledge Graphs and Game-Theoretic Agents

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Abstract

In practice, scenarios are written by managers and policy makers as narratives, and strategy is tested against them. This paper proposes a computational framework that works with those assigned narratives rather than replacing them. It compiles each narrative into a formal specification, tests it for consistency, plausibility and strategic coherence, simulates actor behavior within it, and evaluates candidate strategies across the set.

The world is a sparse temporal knowledge graph (TKG), updated from current reporting by a large language model (LLM) treated as a noisy sensor. A narrative is handled differently, because it is a stipulation about the future rather than evidence about the present. Its premises enter as causal interventions, its actor dispositions as restrictions on hidden preferences, and its end states as conditioning events. Conditioning on end states yields a backcast: the hidden assumptions that must hold for the narrative to come true.

Within each scenario, actor behavior comes from a behavioral sequence model and from approximate equilibria (policy-space response oracles and quantal response), with exploitability reported. A deviation test flags narrated actions that no admissible actor would rationally choose. Strategies are compared in a strategy $\times$ scenario robustness matrix without imposing scenario probabilities, and results return to decision makers as narratives in which every sentence cites the simulated trajectory.

A particle filter tracks which assigned scenario the world is moving toward and identifies the most diagnostic signposts. A sparse Kolmogorov–Arnold network (KAN) provides an approximate, fidelity-scored explanation of the recommended policy. This is a methods proposal; Section 10 specifies its evaluation, and no empirical results are claimed.

1. Introduction

In practice, scenarios are assigned. Managers and policy makers, often with a scenario team, write a small set of narratives, typically three to five, and use them to test strategy and structure debate (Wack, 1985; van der Heijden, 1996). The narrative form is deliberate: it carries causal logic, actor motives and texture that decision makers can discuss and remember.

Narratives also have well-known weaknesses. Their internal consistency is hard to check, the assumptions they depend on are often implicit, and actors in them may behave against their own interests. Testing strategies against them is usually informal, and it is hard to tell which scenario the world is moving toward.

This paper keeps the narratives, and the people who own them, at the center. The proposed pipeline compiles each assigned narrative into a formal specification, tests it, simulates it and reports back in narrative form. The model's job is to reveal tensions and consequences, not to decide what the scenarios should be.

The work draws on established methods. Cross-impact balance analysis checks the consistency of qualitative scenarios (Weimer-Jehle, 2006), and backcasting reasons from a desired end state to the conditions that produce it (Robinson, 1990). Exploratory modeling and Robust Decision Making evaluate strategies across ensembles of futures (Lempert et al., 2003; Bryant & Lempert, 2010), and TKG forecasting predicts events from coded event data (Jin et al., 2020; Liu et al., 2022).

Five commitments shape the design:

1. **The owner decides what a scenario means.** The model translates each narrative, and its owner signs off on the translation.
2. **Stipulation is not evidence.** A narrative about the future changes the model's assumptions; it never updates beliefs about the present.
3. **Findings, not verdicts.** Inconsistencies, hidden assumptions and implausible actor behavior go back to the owner, who may revise the scenario or keep it as written.
4. **Every approximation reports its error.** Extraction accuracy, equilibrium exploitability and surrogate fidelity accompany every result.
5. **Claims are tested out of sample.** The framework is only as credible as its evaluation record.

Section 2 defines the world model and Section 3 how it is updated from current reporting. Sections 4 and 5 compile and test the assigned narratives, Section 6 monitors them, Section 7 models the actors, Section 8 evaluates strategies and Section 9 explains them. Sections 10 and 11 cover evaluation and limitations.

2. Problem Formulation

The world is modeled as a Bayesian partially observable stochastic game played on a sparse, weighted temporal knowledge graph. The main hidden variable is what other actors want, not which facts hold.

2.1 Sparse temporal knowledge graph

Let $\mathcal{E}_t$ be the entity set at time t (size N), $\mathcal{R}$ the typed relation set (size M), and $t \in \{0, 1, \dots, H\}$ discrete time. The graph is a set of weighted facts:

$$\mathcal{G}_t = (\mathcal{E}_t, \mathcal{F}_t), \qquad \mathcal{F}_t \subseteq \mathcal{E}_t \times \mathcal{R} \times \mathcal{E}_t \times \mathbb{R}, \qquad |\mathcal{F}_t| \ll N^2 M$$

A fact (i, k, j, w) states that relation k holds from entity i to entity j with intensity w, such as a 25% tariff. The dense $N \times M \times N$ tensor is a valid notation but is never materialized; all computation scales with the number of facts. The entity set may grow, with new entities admitted only after review (Section 3).

An ontology constrains each relation's domain, range and units. For example, a tariff runs from a state to a (state, sector) pair and carries a rate in percent.

2.2 The game

$$\mathcal{M} = \langle \mathcal{I},\ \mathcal{S},\ \{\mathcal{A}_i\},\ P_\psi,\ \{u_i(\cdot\,; \theta_i)\},\ p(\theta),\ \gamma,\ H \rangle$$

- **Players $\mathcal{I}$.** A small set of strategic actors, typically three to six. All other entities follow exogenous stochastic dynamics. Keeping the strategic set small is a deliberate tractability choice.
- **State $\mathcal{S}$.** Each state $s_t = (\mathcal{G}_t, x_t)$ combines the graph with exogenous indicators x_t such as growth or commodity prices.
- **Actions $\mathcal{A}_i$.** Finite, typed graph edits: impose, raise, lower or remove a tariff; join or exit an agreement; sanction; or hold.
- **Transitions P_ψ.** A stochastic simulator applies the edits and propagates shocks along graph edges. Its parameters ψ are calibrated to history by simulation-based inference (Cranmer et al., 2020).
- **Utilities.** Each actor's payoff is linear in a small set of interpretable features, with private weights θ_i (its type; Harsanyi, 1967):

$$u_i(s; \theta_i) = \theta_i^\top \varphi_i(s), \qquad \theta_i \sim p(\theta_i)$$

Features φ_i might include output growth, a domestic-approval proxy, alliance cohesion and security exposure. The prior $p(\theta)$ comes from structured expert elicitation (Cooke, 1991).

2.3 What is hidden, and the solution concept

Separating the two sources of uncertainty is what makes inference tractable. Facts about the graph are observed noisily through text (Section 3). Actor types θ are never observed and must be inferred from behavior (Section 6).

Policies are behavioral strategies $\pi_i(a \mid h_i, \theta_i)$ over an actor's history h_i. The target is an ε-Bayes–Nash equilibrium, where no actor can gain more than ε by deviating:

$$\mathbb{E}_\theta\big[U_i(\pi_i, \pi_{-i})\big] \;\geq\; \max_{\pi_i'} \; \mathbb{E}_\theta\big[U_i(\pi_i', \pi_{-i})\big] - \varepsilon \quad \forall i \in \mathcal{I}$$

Stronger refinements, such as perfect Bayesian equilibrium, are not claimed. Enforcing sequential rationality with consistent beliefs is intractable at this scale.

3. Present-State Extraction: The LLM as a Noisy Sensor

The LLM is a sensor with measured error rates, not a compiler of ground truth. Its errors enter the belief update explicitly instead of silently fixing the initial conditions.

This section covers reporting about the present: news, filings, intelligence reports. Assigned scenario narratives describe the future and are handled in Section 4; they never enter the belief update as evidence.

3.1 Structured extraction

For each document d, the LLM emits claims $c = (i, k, j, \hat{w}, t, \text{span}, \text{source})$ under constrained decoding against the ontology schema. Entities are linked to canonical identifiers, and each claim must cite a verbatim evidence span or it is rejected. Proposed new entities and claims with high decision impact go to human review.

3.2 Observation model

For each relation type k and source class, error rates are estimated on an annotated validation set: sensitivity r_k (a true fact yields a claim) and false-positive rate f_k (a false triple yields a claim). Reported intensities are noisy, with $\hat{w} \sim \mathcal{N}(w, \sigma_k^2)$.

Absence of a report is evidence only where a source would have reported the fact. Let $\mathcal{C}_t$ be the set of triples the period's documents cover. The likelihood of the observed claims o_t given a hypothesized state is:

$$p(o_t \mid s_t) = \prod_{c \in o_t} p(c \mid s_t) \prod_{\substack{(i,k,j) \in \mathcal{C}_t \cap \mathcal{F}_t \\ \text{unclaimed}}} (1 - r_k)$$

$$p(c \mid s_t) = \begin{cases} r_k \, \mathcal{N}(\hat{w}_c;\, w_{ikj},\, \sigma_k^2) & (i,k,j) \in \mathcal{F}_t \\ f_k \, g_k(\hat{w}_c) & \text{otherwise} \end{cases}$$

Here g_k is the background distribution of intensities reported for false claims. The coverage term $\mathcal{C}_t$ prevents the model from treating every unreported fact as disconfirmed.

3.3 Why this matters

The downstream simulation is stochastic and conditioned on uncertain inputs; it is not deterministic. Extraction quality is reported as per-relation precision, recall and F1 against gold annotations. Source reliability can be set per outlet, which partially addresses deliberate deception in the text record (Section 11).

4. Compiling Assigned Narrative Scenarios

An assigned scenario is a stipulation about the future made by its owner, not evidence about the present. It is compiled into a formal specification that changes the model's assumptions, and its owner approves the compiled version before any analysis runs.

4.1 What a narrative contains

Narrative statements play different formal roles, and conflating them is the central error to avoid. The compiler classifies each sentence by type:

Statement type	Example	Formal role
Premise	"A blockade of the Taiwan Strait begins in mid-2027."	Intervention: time-stamped graph edits imposed with the do-operator
Driver assumption	"Oil stays above $100 a barrel through 2029."	Constraint on the exogenous shock path ξ
Actor disposition	"Beijing puts self-sufficiency ahead of growth."	Restriction of the type prior $p(\theta)$ to a region Θ_n
Structural change	"A Gulf–Asia payment system emerges."	Ontology extension: a new entity or relation, reviewed
Milestone or end state	"By 2032 the EU sources under 10% of its chips from Asia."	Conditioning event on trajectories, hard or soft
Causal claim	"Because of the blockade, insurers withdraw from the region."	Hypothesis tested in Section 5, not imposed

The distinction between premises and end states matters most (Pearl, 2009). A premise is imposed without asking why it happened. An end state is conditioned on, so the model must find the paths and hidden conditions that lead to it, which is backcasting (Robinson, 1990).

4.2 The scenario specification

Each narrative n compiles to a specification:

$$\Sigma_n = \langle\, \mathcal{D}_n,\ \mathcal{X}_n,\ \Theta_n,\ \mathcal{O}_n,\ \mathcal{C}_n \,\rangle$$

$\mathcal{D}_n$ holds the time-stamped interventions, $\mathcal{X}_n$ the constraints on drivers, Θ_n the admissible actor types, $\mathcal{O}_n$ the ontology extensions and $\mathcal{C}_n$ the conditioning events. The scenario's distribution over trajectories τ is then:

$$p_n(\tau) \ \propto\ p(\tau \mid \mathrm{do}(\mathcal{D}_n);\ \xi \in \mathcal{X}_n,\ \theta \in \Theta_n,\ \mathcal{O}_n) \prod_{c \in \mathcal{C}_n} \ell_c(\tau)$$

$$\ell_c(\tau) = \begin{cases} \mathbf{1}[\tau \vDash c] & c \text{ hard} \\ \exp(-\kappa_c\, \mathrm{viol}_c(\tau)) & c \text{ soft} \end{cases}$$

Everything the narrative leaves unsaid stays uncertain, drawn from the model's current beliefs (Section 6). A scenario is therefore a distribution of futures consistent with the narrative, not a single path.

4.3 Compilation procedure

1. **Extract.** An LLM with schema-constrained decoding maps each sentence to specification elements. Every element links back to the sentence it came from.
2. **Flag what does not fit.** Sentences the ontology cannot express are listed as outside the model, never silently dropped.
3. **Resolve vagueness.** Terms like "significant tariffs" or "within a few years" are offered to the owner as ranges to choose from. Terms left unresolved become distributions and are flagged.
4. **Round-trip.** The specification is rendered back into prose, and the owner compares it with the original and corrects any drift.
5. **Sign off.** The owner approves the specification. The model never edits a scenario; it only reports on it (Section 5).

5. Testing Assigned Scenarios

Each approved specification is tested on five properties before any strategy is evaluated, and the set is then checked as a whole. The results go back to the owner as findings; the owner decides whether to revise the scenario or keep it as written.

5.1 Consistency

Hard statements are checked for joint satisfiability as constraints over the specification. For example, "tariffs are removed in 2028" contradicts "tariffs stay above 30% throughout the decade." This mechanizes what cross-impact balance analysis does by hand for qualitative scenarios (Weimer-Jehle, 2006).

5.2 Plausibility given the premises

The test estimates how likely the narrative's end states are once its premises are imposed:

$$\rho_n = P(\mathcal{C}_n \mid \mathrm{do}(\mathcal{D}_n),\ \mathcal{X}_n,\ \Theta_n)$$

Rare end states are estimated with sequential importance sampling rather than naive simulation. ρ_n is not the probability of the scenario, since its premises are stipulated. It measures how far the narrative departs from the model's dynamics; a low value is normal for a deliberately stretching scenario, but the owner should know it.

5.3 Backcast: what must be true

Conditioning on the end states shifts beliefs about everything the narrative left unsaid. Comparing that posterior with the prior exposes the narrative's hidden assumptions:

$$p(\theta, \xi, \psi \mid \mathrm{do}(\mathcal{D}_n),\ \mathcal{C}_n) \quad \text{vs.} \quad p(\theta, \xi, \psi)$$

The variables with the largest shifts are reported in plain language. An example finding: "This scenario requires actor B to weight security above output by more than 2 to 1, which the elicited prior rates at about 12%." The typical paths that reach the end state are reported too, including when each milestone tends to occur.

5.4 Strategic coherence

Narratives often have actors make choices. For each narrated choice $a_{j,t}$, the test computes actor j's best gain from deviating, under every admissible type:

$$G_{j,t} = \min_{\theta_j \in \Theta_n} \Big[\max_{a'} Q_j(h_t, a';\, \theta_j) - Q_j(h_t, a_{j,t};\, \theta_j) \Big]$$

If $G_{j,t}$ is large, no admissible motive makes the choice rational. The owner then has three options: widen the actor's disposition, keep the choice as a deliberate miscalculation (an intervention), or revise the narrative. By default, choices stated as premises are interventions, and all other actors' reactions are modeled endogenously.

5.5 Causal claims

Each narrated causal link is tested by comparing the model with and without its cause, $\mathrm{do}(B)$ against $\mathrm{do}(\neg B)$. The report states whether the model reproduces the effect and how large it is. Disagreement is a finding about the narrative or the model; a missing mechanism may call for ontology review.

5.6 The scenario set as a whole

Each scenario is mapped onto the main uncertainty dimensions (drivers and actor types). The report flags pairs whose outcome distributions overlap heavily, and large plausible regions that no scenario covers. Blind spots are offered to the owners as candidate additions, never added automatically.

6. Belief Tracking and Scenario Monitoring

Beliefs are represented by K weighted particles over the joint latent $z_t = (s_t, \theta, \psi)$, not by a distribution over an enumerated state simplex. Each particle is one complete hypothesis about the world, the actors' preferences and the simulator's parameters.

6.1 Update

Each particle is propagated through the simulator using the actions observed in period t. Its weight is then updated by the text evidence (Section 3) and by how well its hypothesized types explain the actions actors actually took:

$$w_t^{(k)} \;\propto\; w_{t-1}^{(k)}\; p(o_t \mid s_t^{(k)}) \prod_{j \in \mathcal{J}} \pi_j(a_{j,t} \mid s_{t-1}^{(k)},\, \theta_j^{(k)})$$

The action term is a Bayesian inverse reinforcement learning step (Ramachandran & Amir, 2007). For speed, filtering uses a myopic logit choice model:

$$\pi_j(a \mid s, \theta_j) \;\propto\; \exp(\lambda\, \hat{Q}_j(s, a;\, \theta_j))$$

$\hat{Q}_j$ is a short simulator lookahead under θ_j, and λ is a rationality parameter. This is an approximation, adopted because solving the full game per particle per step is infeasible.

6.2 Degeneracy and cost

Static parameters such as θ and ψ collapse onto a few values in a plain particle filter. Particles are therefore resampled when the effective sample size falls below $K/2$, then rejuvenated with an MCMC move on the static parameters (Gilks & Berzuini, 2001).

Because the graph is sparse, evaluating the likelihood costs $O\big(K(|\mathcal{F}_t| + |o_t|)\big)$ per step, plus K simulator steps. Nothing scales with $N^2 M$.

6.3 Which assigned scenario is unfolding?

The same filter, run under each scenario's specification, shows which narrative the evidence favors. Its unnormalized weights give the marginal likelihood of the evidence so far under scenario n:

$$\hat{p}(o_{1:t} \mid \Sigma_n) = \prod_{s=1}^{t} \frac{1}{K} \sum_{k=1}^{K} \tilde{w}_s^{(k)}$$

Ratios of these (Bayes factors) show which assigned scenario the world is tracking. They are relative: every scenario may fit poorly, so each is also compared with the unconstrained model. Before a premise's date arrives, evidence can only bear on its precursors.

Signposts are the observable variables whose predicted values over the next 6–12 months differ most between scenarios. These are the most diagnostic indicators to watch, and each is tied to the strategy triggers in Section 8.3.

7. Actor Models: Behavioral Priors and Approximate Equilibria

Within each compiled scenario, actor behavior is generated by two complementary models, because neither is adequate alone. A behavioral model says what actors like this have done; an equilibrium model says what rational actors with these preferences would do.

7.1 Behavioral prior

A sequence model (a Decision Transformer or a TKG forecaster) is trained on historical event sequences. States are embedded with a message-passing graph encoder whose cost is linear in $|\mathcal{F}_t|$. This model imitates the data; it does not optimize and carries no equilibrium guarantee (Chen et al., 2021).

If return-conditioning is used, target returns are restricted to the support of the training data. Vanilla return-conditioning is biased in stochastic environments, because it credits luck as skill; targets are therefore averaged over clusters of similar trajectories (Paster et al., 2022).

7.2 Approximate equilibria with PSRO

Policy-space response oracles (Lanctot et al., 2017) maintain a population of policies for each actor, initialized with behavioral-prior policies so equilibria stay anchored to realistic behavior. Each iteration:

1. Simulate every combination of population policies to fill an empirical meta-game payoff table (Wellman, 2006).
2. Solve the meta-game for a mixture over each actor's population.
3. Train an approximate best response for each actor by deep RL in the simulator, and add it to that actor's population.
4. Stop when approximate best responses no longer improve payoffs by more than a threshold.

Each result is reported with its exploitability, the total gain actors could obtain by unilateral deviation:

$$\text{NashConv}(\pi) = \sum_{i \in \mathcal{J}} \left[\max_{\pi_i'} U_i(\pi_i', \pi_{-i}) - U_i(\pi) \right]$$

The maximum is approximated by trained best responses, so the reported value is a lower bound. True exploitability may be higher, and results say so.

7.3 Bounded rationality and equilibrium selection

Computing Nash equilibria is PPAD-complete even for two-player general-sum games (Daskalakis et al., 2009; Chen et al., 2009). Approximation is unavoidable, and games often have several equilibria.

The meta-game is also solved for a logit quantal response equilibrium (McKelvey & Palfrey, 1995):

$$\sigma_i(\pi_i) = \frac{\exp(\lambda\, U_i(\pi_i, \sigma_{-i}))}{\sum_{\pi_i'} \exp(\lambda\, U_i(\pi_i', \sigma_{-i}))}$$

At $\lambda = 0$ play is uniform; as λ grows, play approaches Nash. For generic games, following the principal branch from $\lambda = 0$ selects a unique limiting equilibrium. λ is estimated from history and sampled as an uncertain parameter in the ensemble.

When different solver seeds converge to distinct equilibria, each is kept as a separate branch of the ensemble. Branches are never averaged into a single "expected" equilibrium.

7.4 Cross-actor relevance

Attention weights are not treated as explanations (Jain & Wallace, 2019). The relevance of actor j to actor i is measured by intervention: replace j's policy with a baseline, such as the status quo, and record the change in i's payoff.

$$\Delta_{ij} = U_i(\pi) - U_i(\pi_i,\, \pi_j^{\text{base}},\, \pi_{-ij})$$

8. Evaluating Strategies Across Assigned Scenarios

Strategies are evaluated inside each assigned scenario and compared across the set without assigning probabilities to scenarios. This matches how scenario planning is used: to find strategies that hold up across futures the organization has agreed to take seriously.

8.1 Within-scenario evaluation

Candidate strategies $\mathcal{D}$ for the focal actor come from decision makers, plus options from the actor's PSRO population, always including the status quo. For each scenario n, worlds ω are sampled from p_n (Section 4.2), covering types, drivers, rationality λ and equilibrium branch. The focal actor commits to $d \in \mathcal{D}$, other actors respond according to Section 7, and the focal actor receives value $V(d, \omega)$.

8.2 The robustness matrix

Results form a strategy $\times$ scenario matrix. Each cell reports the median value, a downside value (the 10th percentile) and regret against the best candidate in that scenario:

$$\mathrm{Reg}_n(d) = \max_{d' \in \mathcal{D}} \mathbb{E}_{\omega \sim p_n}\big[V(d', \omega)\big] - \mathbb{E}_{\omega \sim p_n}\big[V(d, \omega)\big]$$

Across scenarios, the report shows each strategy's worst-case regret and flags strategies that are dominated in every scenario. It does not weight scenarios by default. If decision makers choose weights, the weights are theirs and the sensitivity of the ranking to them is shown.

8.3 Adaptive strategies

The signposts of Section 6.3 become triggers. An adaptive strategy has the form "start with d_1; switch to d_2 if the Bayes factor for scenario B exceeds 10." Adaptive strategies are evaluated in every scenario like any other candidate, including the cost of switching late.

8.4 Reporting in narrative form

Results go back to decision makers as narratives, the medium they work in. For each scenario and strategy, the LLM narrates a median trajectory and a downside trajectory. Every sentence must cite the simulated events it describes; sentences without a citation are rejected.

Each report also states where the simulation departs from the assigned narrative. An example: "In the model, the blockade ends in 2029 under pressure from Asian importers, rather than escalating as the narrative describes."

8.5 Blind-spot check

As a secondary step, scenario discovery searches the unconstrained ensemble for conditions outside the assigned set under which the preferred strategy fails (Bryant & Lempert, 2010; Friedman & Fisher, 1999). Each finding is presented as a short candidate narrative. Owners may adopt it, rewrite it or set it aside.

9. Interpretable Policy Surrogates

A sparse KAN approximates the recommended policy over a small set of expert-defined concepts, and every extracted formula ships with its measured fidelity. The explanation describes the surrogate, not the underlying model, and says so.

9.1 Concept layer

Raw inputs of dimension N^2M cannot yield a readable formula. Experts instead define $p \approx 5$–20 concepts $g(s) \in \mathbb{R}^p$, such as trade exposure to actor B, B's average tariff level, an alliance cohesion index and months to B's next election. Each concept is an explicit, auditable function of graph facts.

9.2 Surrogate architecture

The KAN maps $(g(s), \tau)$ to action logits, where τ is time to horizon. Time enters as an ordinary input rather than through drifting spline coefficients, so each edge stays a fixed univariate function and the extracted expression is static. Each layer follows Liu et al. (2024):

$$x_{l+1,\,j} = \sum_{i=1}^{n_l} \phi_{l,j,i}(x_{l,i}), \qquad \phi(x) = w_b\,\mathrm{silu}(x) + w_s \sum_{m=1}^{G+k} c_m B_m(x)$$

9.3 Training and sparsification

Training states are drawn from the ensemble's visitation distribution under the recommended strategy profile, plus perturbed off-distribution states to test extrapolation. The loss matches the target policy with an L_1 penalty on edge activations:

$$\mathcal{L} = \mathbb{E}_s\Big[\mathrm{KL}\big(\pi^*(\cdot \mid s)\,\big\|\,\widehat{\pi}(\cdot \mid s)\big)\Big] + \lambda_1 \sum_{l,i,j} \big\|\phi_{l,j,i}\big\|_1$$

The penalty weight λ_1 is chosen on a validation fidelity–complexity frontier. It is not driven to infinity: in that limit every

edge is pruned and the surrogate becomes a constant.

9.4 Symbolic snapping

Each surviving edge is fitted against a library of simple functions (polynomial, exponential, logarithm, sigmoid, step). Edges that fit above an R^2 threshold are replaced by their formula. Edges that do not are left as splines and shown as plots, rather than forced into a misleading expression.

9.5 Reporting and baselines

Fidelity is reported as action agreement and KL divergence on held-out in-distribution and off-distribution states. Complexity is reported as active edges and expression size. Regions where agreement falls below a threshold, such as 90%, are flagged as not reliably explained.

Strategic policies are often threshold-like, which splines represent poorly. The KAN is therefore compared at equal complexity with optimal sparse decision trees (Hu et al., 2019), PySR symbolic regression (Cranmer, 2023) and logistic regression on the concepts.

Finally, the best surrogate is evaluated as a policy in its own right across the ensemble. If its regret is close to that of the original strategy, the interpretable policy itself should be recommended (Rudin, 2019).

10. Evaluation Protocol

The framework is judged on whether it helps decision makers work with their assigned scenarios: faithful translation, useful test findings, accurate monitoring and better strategy discussions. Strategy quality cannot be validated directly, because counterfactual outcomes are never observed.

10.1 Archived scenario sets

Published scenario sets written years before the events they describe provide a natural test bed. Examples include past editions of the US National Intelligence Council's Global Trends reports and corporate scenario sets. Each set is compiled and tested as if new.

The key question is whether the tests would have flagged the elements that later failed. These include inconsistent premises, implausible end states, incoherent actor behavior and unsupported causal claims. Monitoring is checked by replaying the evidence that followed and asking whether the Bayes factors favored the closest-matching scenario earlier than contemporaneous analysts did.

10.2 Leakage control

Pretrained LLMs have usually read about the outcomes, which contaminates any retrospective test they touch. Three mitigations are used together: models with training cutoffs before the scenario date where available, anonymized entity names, and prospective evaluation on scenario sets written after the model's cutoff.

10.3 Metrics and baselines

Stage	Metric	Baseline
Narrative compilation	Element-level agreement with independent expert encodings; owner acceptance after round trip; share of sentences flagged outside the model	Expert encoding alone
Scenario tests	Detection rate on narratives seeded with known contradictions or incoherent behavior; false-alarm rate on vetted narratives	Cross-impact balance analysis by hand
Present-state extraction	Precision, recall and F1 per relation type	Existing coded event data (ICEWS, GDELT)
Forecasts and monitoring	Brier and log scores; calibration; lead time in identifying the closest scenario	Base rates, persistence, RE-NET, TLogic, analyst judgments
Equilibria	NashConv lower bound; stability across seeds	Behavioral prior alone
Surrogates	Fidelity and complexity (Section 9.5)	Sparse trees, PySR, logistic regression
Decision support	Hidden assumptions and failure conditions identified; analyst calibration; time taken	Scenario workshop without the tool

10.4 Ablations

Each commitment is tested by removing it. Ablations include treating narrative statements as evidence about the present, imposing end states as interventions instead of conditioning on them, skipping the strategic coherence test, and using only the behavioral or only the equilibrium model. Compute cost is reported for every configuration.

11. Limitations and Responsible Use

The framework's outputs are conditional on its model of the world, and several limits cannot be engineered away.

- **Translation loss.** Narratives carry meaning, tone and deliberate ambiguity that no specification captures. Unmapped sentences are listed, but nuance is still lost; the narrative, not its specification, remains the authoritative document.
- **Model authority.** Test findings reflect the model, not reality. A scenario flagged as implausible may reveal a missing mechanism rather than a flawed narrative, so findings inform the owner's judgment and never reject a scenario automatically.
- **Anchoring.** Showing plausibility scores may push owners toward conventional scenarios and away from the stretching ones that make scenario work valuable. Scores should be framed as departures from the model, not as grades.
- **Deep uncertainty.** Outside the assigned narratives, the model spans only what its ontology and priors can express. New actors, technologies and regime changes may fall outside it, and every output is conditional on the model.
- **Behavioral assumptions.** Linear utilities over a few features, and logit rationality, are strong simplifications of real political decision-making.
- **Deception and poisoning.** Adversaries shape the reporting the system reads. Provenance, per-source reliability and human review reduce this risk but do not remove it.
- **Reflexivity.** Forecasts and signposts that are acted on, or leaked, change the behavior they model.
- **Small samples.** Few archived scenario sets and historical episodes are comparable, so evaluation results will carry wide intervals.
- **Computational cost.** Conditioning on rare end states and running PSRO in every scenario is expensive, which is why the strategic player set is kept small.
- **Automation bias and accountability.** Precise-looking output invites over-trust. Uncertainty must be shown prominently, decisions stay with named human owners, and every run is logged for audit.
- **Dual use.** The same machinery can support coercive strategy, so deployment should sit under explicit institutional governance.

12. Conclusion

Scenarios belong to the people who write and use them. This framework accepts assigned narratives as its input and does what workshops find hard: it checks their consistency, exposes the assumptions they depend on, tests whether their actors behave coherently, simulates their consequences and compares strategies across them.

The key design choice is to treat a narrative as a stipulation, not evidence. Premises are imposed, end states are backcast, and the results return as cited narratives that owners can accept, question or revise. Whether this improves strategic decisions over a well-run workshop is an empirical question, and Section 10 sets out how to answer it.

Appendix A: Changes from the Original Draft

Every equation is now stated in full; the original's display equations were blank. The table maps each substantive claim in the original to its problem and its revision.

Original claim	Problem	Revision	Section
An LLM maps qualitative scenarios directly into graph mutations	Treats a stipulated future as present fact; conflates premises, dispositions, end states and causal claims; no owner check	Narratives compiled into a typed specification (interventions, constraints, conditioning events), with round-trip owner sign-off	4
Scenarios are paths generated by rational actors	Asserted, never tested	Strategic coherence test with deviation gains; incoherent choices reported to the owner	5.4

Original claim	Problem	Revision	Section
The strategist's task is one optimal trajectory	Assigned scenarios are plural and unweighted	Strategy × scenario robustness matrix without scenario probabilities; adaptive strategies	8
Output is an algebraic formula for policy makers	Decision makers work in narratives	Results returned as narratives citing simulated events, plus fidelity-scored surrogates	8.4, 9
Belief over a simplex of states in $\mathbb{R}^{N^2M}$	A simplex needs a finite state set; even the binary graph has 2^{N^2M} states	Particles over a sparse graph plus actor types	2, 6
Hidden utilities cause partial observability, but the belief is over graph state	Uncertainty placed in the wrong variable	Types θ inferred from behavior; graph facts observed through a noise model	2.3, 3, 6
LLM output keeps the simulation "deterministic"	Extraction errors silently fix the initial conditions	Calibrated noisy-sensor likelihood for present-state reporting	3
A Decision Transformer computes Nash equilibria	DTs imitate data and carry no equilibrium guarantee	DT kept as a behavioral prior; equilibria from PSRO	7.1, 7.2
The trajectory is a "true equilibrium"	PPAD-hardness and multiple equilibria	ε-Bayes–Nash with NashConv lower bound; QRE selection; branches kept separate	7.2, 7.3
Attention weights form an adjacency matrix of strategic relevance	Attention is not a reliable explanation	Interventional relevance Δ_{ij}	7.4
KAN on the raw state; $\lambda \to \infty$ prunes redundant edges	Unreadable at scale; the limit is a constant function	Concept layer, validated λ_1, fidelity reporting, tree and PySR baselines	9
Time-driven spline coefficients	Extracted expression is not static	Time as an ordinary input	9.2
Output is "immediately actionable"	No evidence offered	Evaluation on archived scenario sets, with leakage controls and explicit limitations	10, 11

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